

Dinul Iman

Full-stack Developer · Jakarta, Indonesia

Full-stack developer focused on the backend, with a love for working with data — from payroll systems to ETL pipelines.

✉ dinulimankappa@proton.me · GitHub · LinkedIn

ABOUT

Hi! I'm Dinul Iman, an Information Technology graduate from Universitas Sumatera Utara focused on full-stack development, with my strongest experience in backend and databases. Day to day I work with PHP, TypeScript, Node.js and relational databases such as PostgreSQL and Oracle.

My most notable work is the Payroll Management System I worked on for PT PELNI as a programmer at PT Solusi Integrasi Utama, a PELNI vendor. It serves around 4,800–5,000 employees — from PPh 21 income tax, BPJS, pension and cooperative deductions to journal integration with Oracle. A favourite achievement: cutting the payroll run from more than 10 minutes to under 1 minute by optimizing stored procedures.

I enjoy work that sits close to data: ETL/ELT, data warehousing and automation. Lately I've also been actively exploring AI coding agents such as Claude Code, Codex and MCP. Outside work I build open portfolio projects: document extraction and Text-to-SQL with local LLMs, a Midtrans checkout that survives misbehaving webhooks, and data pipelines with Airflow, Kafka and Spark.

TARGET ROLES

Backend Developer / Engineer · **Full-stack Developer** · **Software Engineer** · **Data Engineer** · **HRIS / Payroll Technology** · **Freelance: backend, data & AI automation**

EXPERIENCE

Programmer — PT Solusi Integrasi Utama (PT PELNI vendor) Aug 2025 — Present

- PT PELNI Payroll Management System (±4,800–5,000 employees): development with PHP/CodeIgniter and PostgreSQL.
- Payroll calculation and processing: PPh 21 income tax, BPJS, pension, cooperative, receivables and salary proration.
- Worked on the data integration Oracle → PostgreSQL → payroll → Oracle journal.
- Optimized payroll stored procedures, cutting a run that took >10 minutes to <1 minute.
- Built payslips and payroll reports with JasperReports/JasperServer.
- PT PELNI ticketing application: backend development with NestJS and PostgreSQL, including the shipping insurance module.
- Used Docker / Rancher Desktop for application environments.

Data Structures & Algorithms Lab Assistant — Universitas Sumatera Utara

Jan 2021 — Jun 2021

- Teaching assistant for the Data Structures & Algorithms lab course.

Database Lab Assistant — Universitas Sumatera Utara Jun 2020 — Jan 2021

- Teaching assistant for the Database lab course.

EDUCATION

B.Sc. Information Technology — Universitas Sumatera Utara (USU)
GPA 3.50 · Lab assistant for Databases and for Data Structures & Algorithms.

SKILLS

PHP & CodeIgniter	★★★★★	Laravel	★★★★☆
Yii2	★★★☆☆	Node.js & NestJS	★★★★☆
TypeScript	★★★★☆	Python (FastAPI, Django)	★★★☆☆
Go & Gin	★★★☆☆	JavaScript	★★★★☆
React & Next.js	★★★☆☆	PostgreSQL	★★★★★
Oracle	★★★★☆	MySQL	★★★★☆
MongoDB	★★★☆☆	ETL & Data Integration	★★★★☆
JasperReports / JasperServer	★★★★☆	Tableau	★★★☆☆
BigQuery	★★★☆☆	Airflow & dbt	★★★☆☆
Spark & Kafka	★★★☆☆	Docker / Rancher	★★★☆☆
Payroll & HRIS	★★★★★	LLM & AI Integration	★★★☆☆
n8n & Workflow Automation	★★★☆☆	Web Scraping	★★★★☆
Payment Gateway	★★★☆☆		

PROJECTS

Payroll Management System for PT PELNI
Payroll system for ±4,800–5,000 employees: PPh 21 income tax, BPJS, pension, cooperative, receivables and proration; Oracle ↔ PostgreSQL integration; payslips & reports via JasperServer. Stored procedure optimization cut the run from >10 minutes to <1 minute. Built as a programmer at PT Solusi Integrasi Utama.

PHP · CodeIgniter · PostgreSQL · Oracle · JasperServer · Docker

Ticketing Application for PT PELNI
Backend for a ticketing application, including a shipping insurance module. Built at PT Solusi Integrasi Utama.

NestJS · TypeScript · PostgreSQL

Data Engineering Portfolio link

Three end-to-end data pipelines built like small production systems, with CI: batch ELT of weather data for 10 Indonesian cities (Airflow, MinIO, PostgreSQL, dbt, Metabase), real-time e-commerce order streaming (Kafka, Spark Structured Streaming with watermarks, dead-letter queue, Grafana), and an NYC Taxi bronze → silver → gold lakehouse (PySpark, Delta Lake, quality gates, time travel).

Python · Airflow · dbt · Spark · Kafka · Delta Lake · PostgreSQL · Docker

Naver SmartStore Scraper API link

REST API that extracts product data from Naver SmartStore pages using a stealth headless browser, rotating proxies & fingerprints, throttling and caching. The scraping backend can be swapped via an env var with no code changes.

TypeScript · Node.js · Playwright

AI Document Extractor link

Turns invoice PDFs into structured data in PostgreSQL with a local LLM (Ollama), so documents never leave the server. The model only reads; dates and currency are normalised in code, then the numbers are checked (qty × price, subtotal, tax, total). Anything that doesn't add up goes to a review queue. 100% accuracy on the test set (60/60 fields, 20/20 line items).

Python · FastAPI · Ollama · PostgreSQL · Pydantic

Text-to-SQL Agent link

Ask a PostgreSQL database in plain language (English/Indonesian): a local LLM writes the SQL, then a sqlglot guard and a read-only database role make sure it can only read. Results come back as a table and a chart. 20/24 correct on the test set, and 6/6 hostile requests refused without touching the data.

Python · FastAPI · Ollama · sqlglot · PostgreSQL

AI Ticket Triage with n8n link

n8n workflow that reads customer messages (Indonesian/English), classifies category & priority with a local LLM, drafts a reply, stores the ticket in PostgreSQL and alerts Slack/Discord when it's urgent. Plus a daily digest.

n8n · Ollama · PostgreSQL · JavaScript · Docker

Price Tracker link

Async scraper that collects 1,000 products in about 60 seconds with retries and rate limiting, keeps price history in PostgreSQL, detects price/stock changes, sends alerts and shows a Streamlit dashboard.

Python · httpx · PostgreSQL · Streamlit · pytest

PostgreSQL Payroll Optimization Case Study link

A real payroll problem rebuilt with synthetic data for 5,000 employees: from a 28-second row-by-row loop to 0.2 seconds (140x) using indexes, caching outside the loop and a set-based version, with results verified identical row by row.

PostgreSQL · PL/pgSQL · Docker

Leave Management API link

Employee leave REST API in Laravel: working-day calculation (weekends & holidays excluded), per-type balances, pending requests that reserve balance, manager approval, and row locking so parallel requests stay safe. 20 tests run on SQLite and PostgreSQL in CI.

PHP · Laravel · Sanctum · PostgreSQL · PHPUnit · GitHub Actions

GoBack — E-Commerce Platform link

Full-stack e-commerce app: Go backend with JWT auth, product & stock management, addresses, Midtrans payments and an admin dashboard; React frontend.

Go · Gin · GORM · PostgreSQL · React · JWT

Payment Webhooks (Midtrans) in Go link

Midtrans checkout that survives forged, duplicate, out-of-order and lost webhooks: signature and amount checks, database-level idempotency, a status machine and reconciliation via the status API. Ships with a local Midtrans simulator; every scenario runs in CI.

Go · Gin · PostgreSQL · Midtrans · Docker

Job Market Analytics

Job market analysis platform: job-posting scraping, automatic skill extraction, in-demand skills, salary statistics by location & level, and career path recommendations.

Python · Flask · Vue.js

Botto — RAG Chatbot link

Chatbot for asking questions about your own documents (PDF, Word, TXT) using Retrieval-Augmented Generation with streaming responses. The whole stack is free.

Next.js · TypeScript · LangChain.js · ChromaDB · Groq (Llama 3.1)

Toko Syuhada — Bicycle Shop Website link

Bicycle catalog & sales website with category filters, product search, detailed specs and articles.

Next.js · TypeScript · Prisma · PostgreSQL · Tailwind CSS

Machine Learning Experiments link

A collection of notebooks: an LSTM crypto Telegram chatbot (thesis), crypto price forecasting with LSTM, sentiment analysis, a seq2seq chatbot, and data analysis with Pandas and PySpark (car prices, ad campaigns, diabetes).

Python · TensorFlow/Keras · Pandas · PySpark

Apple or Tomato — Real-time Image Classifier link

Tells apples from tomatoes straight from a webcam: MobileNetV2 transfer learning with data augmentation, and an OpenCV app that shows the prediction and its confidence in real time.

Python · TensorFlow/Keras · MobileNetV2 · OpenCV

CV Quest link

An interactive CV as a browser RPG — yes, the game you're playing right now!

JavaScript Canvas HTML/CSS

INTERESTS

AI & Data · ETL/ELT & Data Warehousing · TypeScript · AI Coding Agents · Claude Code · Codex · MCP · Automation